Scoring Library Classification Schemes with Hierarchical-Clustering Objectives: A Survey of Open Schemes and a Common Benchmark

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Abstract

Library classification schemes are compared mostly by description: their history, notation, coverage and licence. Quantitative comparisons exist, but they measure retrieval effectiveness, the scatter of a subject across schemes or coverage of a reference map of knowledge. We found no study that scores the hierarchy of several universal or near-universal schemes on one structural objective over the same set of works. We survey thirteen open or openly obtainable scheme builds and propose such a comparison, built on objectives from the theory of hierarchical clustering. A scheme, restricted to the classes a corpus uses, is a rooted tree over the works. We score that tree with Dasgupta's cost and Moseley and Wang's revenue on a fixed k-nearest-neighbour graph of text embeddings, and normalize revenue against an average-linkage tree on the same graph; the ratio is called skill. The corpus is 10,000 arXiv papers with CC0 metadata plus a holdout of 500 Open Library editions with librarian-assigned LC call numbers. We keep source-assigned ("gold") placements apart from placements made by one lexical classifier. On gold labels, arXiv's own taxonomy reaches skill 0.786 on arXiv papers and LCC reaches 0.419 on the books. With the classifier's placements the same two schemes fall to 0.128 and 0.005. Across 13 schemes placed by the classifier, skill ranges from 0.059 (msc) to 0.423 (openalex-topics), and correlates with coverage at Spearman $\rho = 0.76$. The classifier, not the scheme, accounts for most of the differences in classifier-placed scores. We also report a classifier defect that the benchmark exposed, and a short experiment in fitting an order-preserving scheme to the same objective, whose results are in-sample or share the graph's embedding and are not evidence of generalization.

Keywords: library classification; hierarchical clustering; Dasgupta cost; knowledge organization; open data; benchmark

1 INTRODUCTION

A universal classification scheme gives every work one address in a single tree. The Library of Congress Classification (LCC), the Dewey Decimal Classification (DDC), the Universal Decimal Classification (UDC), the Basic Classification (BK), the Integrative Levels Classification (ILC) and others each claim to arrange all of knowledge. A practitioner choosing among them, or a researcher designing a new one, needs to ask which arrangement keeps related works close together, and at what navigation cost. The literature answers this mostly qualitatively, by notation, facet structure, history and institutional adoption [1], [2], [3], [4]. Quantitative comparisons have measured retrieval performance [5], [6], [7] or the scatter of a subject across schemes [8], [9], not the structure of the hierarchy itself (Section 1.1). A structured search of JASIST, Knowledge Organization, Cataloging & Classification Quarterly, ISKO and SIG/CR proceedings, JCDL and TPDL (project report 8, `research/reports/8-prior-work-search.md`) found no study that scores several universal schemes on a common hierarchical objective over the same corpus, and no use of Dasgupta cost or Moseley–Wang revenue on a human-built scheme. The search covered titles and abstracts, not full texts (Section 9), so this is absence of evidence, not proof.

Three developments make the question more pressing.

1. **The main lookup service is gone.** OCLC Classify returned the most common DDC and LCC numbers for a work from WorldCat holdings. OCLC discontinued it on 31 January 2024 [10]. No free service replaced it.
2. **Automatic placement has not closed the gap.** Annif, the most mature open subject-indexing toolkit, is trained separately for each vocabulary [11]. On more than 230,000 Swedish union-catalogue records its best (ensemble) configuration reaches 66.82% accuracy at three-digit DDC [12]. Recent work that uses large language models (LLMs) for LCC and DDC assignment reports weakness with more granular classes [13] and a lack of specificity [14], and prompted LLMs fall well short of fine-tuned smaller models on DDC classes for research data [15]. The Library of Congress's own experiments predicted subject and genre with under 50% accuracy, short of a 95% automation threshold [16]. Placement is getting cheaper, so the open question shifts towards *which scheme* to place into.
3. **Licences constrain open use.** DDC is copyrighted by OCLC [17]. The full UDC is licensed [18]. Openly licensed universal schemes are shallow, such as BK [19] and the Free Decimal Correspondence (FDC) [20], or their licence terms are unsettled, as for ILC2 [21]. Data-driven vocabularies such as OpenAlex Topics are open but cover research literature only [22]. A quantitative comparison helps decide whether an open scheme is good enough to adopt.

Contributions. (i) A survey of the thirteen scheme builds benchmarked here, with DDC and UDC as context, covering structure, size, maintainer and redistribution status (Section 2). (ii) A scheme-agnostic scoring method. It induces a tree from a scheme and a set of placements, and scores the tree with Dasgupta cost and Moseley–Wang revenue on a fixed similarity graph. The score is normalized against an average-linkage reference tree on the same graph, and coverage, structure and navigation statistics are reported with it (Section 3). (iii) A pinned, CC0-metadata corpus of 10,000 arXiv papers and a 500-edition LC holdout (Section 4). (iv) Results for 13 scheme builds. Gold placements and classifier placements are kept apart, and the classifier bias is calibrated where both exist (Section 5). (v) A methods note on a classifier defect that the benchmark exposed, with its before-and-after effect (Section 6). (vi) A short account of an order-preserving scheme fitted to the same objective, with its caveats (Section 7).

1.1 Related work

Comparing classification schemes. The Cranfield project compared the Universal Decimal Classification, an alphabetical subject catalogue, a faceted classification and Uniterm coordinate indexing, applied to 18,000 aeronautics documents [5], [23]. Keen compared five index languages built for a test collection on retrieval effectiveness [6], and Carpenter et al. compared five patent classifications on recall and precision for the same questions [7]. O'Neill et al. measured how the classes of LCC and DDC disperse into each other on MARC records [8], and Connaway and Sievert compared the scatter of one subject across LCC, DDC and the NLM Classification [9]. Zins and Santos mapped the top two levels of LCC, DDC and UDC onto one reference model of knowledge [2]. In bibliometrics, research classifications are scored on citation-based criteria: algorithmic journal maps against the ISI Subject Categories [24], the Web of Science and Scopus journal classifications against each other [25], and taxonomies built from different citation relations against one gold standard [26]. These studies score schemes by retrieval, by the scatter of a mapping or by citation coherence. None scores the tree itself. For a single vocabulary, corpus-weighted pruning measures how much of a subject tree a collection needs [27], checking functions detect quality defects in SKOS vocabularies [28], and structural analysis is one of the established ways to evaluate a thesaurus [29].

Objectives for hierarchical clustering. Dasgupta defined the cost of a tree T over items with pairwise similarities w_{ij} as $\sum_{i<j} w_{ij}\ |\text{leaves}(T[i \vee j])|$, where $i \vee j$ is the lowest common ancestor (LCA) of i and j [30]. Similar pairs should be separated as deep in the tree as possible. A flat tree has the maximum cost. Moseley and Wang studied the dual, revenue, $\sum w_{ij}(n - |\text{leaves}(T[i \vee j])|)$, so that cost plus revenue equals $n \sum w_{ij}$. They showed that average linkage is a $\frac{1}{3}$-approximation to the maximum revenue [31], [32]. Optimizing Dasgupta cost exactly is NP-hard. Approximation algorithms use sparsest cut and spreading metrics [33], [34]. Cohen-Addad et al. give an axiomatic family of admissible objectives and show when average linkage and its variants recover planted hierarchies [35]. Hyperbolic embeddings give a differentiable relaxation [36], and structural constraints can be added to the search [37]. scikit-network ships a Dasgupta cost and score for binary dendrograms [38]. This line of work treats the objective as something to *optimize*. We use it first to *evaluate* trees that people built.

No objective-free optimum. Kleinberg's impossibility theorem says no clustering function satisfies scale invariance, richness and consistency together [39]. Carlsson and Mémoli characterize single linkage axiomatically among hierarchical methods [40]. One consequence for us: a score is always relative to the similarity function. We therefore fix and publish the graph and treat it as part of the measurement.

Navigation. Optimal search trees and Huffman coding minimize expected depth under an access distribution [41], [42]. The hierarchical-identity model of Watts, Dodds and Newman makes greedy navigation on a hierarchy a measurable success criterion [43]. Basic-level categories explain why a natural middle level carries most human use [44]. User studies of menu hierarchies disagree on the best shape. Speed and accuracy improved with breadth in one study [45], and task time rose with depth for web links [46], but others found an optimum at moderate breadth [47], [48]. Landauer and Nachbar fit the time per menu screen as $k + c \log b$ for b choices [49], the form of our $\text{nav}_{\log}$ model. For a two-level card file, the optimum number of records per guide is proportional to the square root of the file size [50]. We report simple, uniform-use navigation costs and fit no user model.

Data-driven and LLM-built taxonomies. Waltman and van Eck cluster publications on direct citation at several resolutions [51], using the Leiden algorithm [52]. OpenAlex Topics come from citation clustering and are named by an LLM, and their upper levels are mapped from Scopus ASJC categories [22], [53]. TaxoGen and TaxoExpan build and extend topic taxonomies from text [54], [55]. LLM pipelines generate layers or labels: Chain-of-Layer, TnT-LLM and TaxoAdapt [56], [57], [58]. None of these optimizes a global objective. Taxonomy-quality work catalogues attributes and measures [59]. It includes a reference-free robustness score that correlates semantic and taxonomic similarity [60], which is closest in spirit to ours.

Automatic subject classification. Annif [11], its SemEval-2025 Task 5 entry [61] and the task overview [62] cover subject indexing with thesauri such as GND. They report results per vocabulary, not across schemes. An open LCC classifier covers 15 of the 21 top-level classes [63]. We found no public leaderboard that compares classification schemes, as opposed to classifiers.

Table 1. The classification systems: thirteen benchmarked builds, with DDC and UDC as context. "Classes" is the number of classes in the build used in the benchmark. "Bundled" means the data file is committed in the repository.

scheme	maintainer	version	classes	structure	licence and redistribution status
`lcc`	Library of Congress	2024	7,392	21 main classes (letters); subclasses and numbered spans (outline)	US government work, 17 U.S.C. 105; LC's Classification Web notice calls LC classification copyrighted outside the USA. Bundled
`lcc-full`	Library of Congress	2024	416,186	13 levels; 408,794 numbered entries from 44 FreeLCC text PDFs added to the outline; max 7,994 children	As `lcc`. Bundled
`fdc`	J. M. Ockerbloom	0.06 (2010-08-19)	1,097	6 levels; Dewey-shaped numbers 000 to 999.9999; written from DDC 11th ed. (1922)	CC0 1.0. Frozen; author states no long-term maintenance. Bundled
`bk`	GBV (K10plus)	export 2026-10-03	2,140	Five roots, two-digit headings, decimal classes; academic bias	CC0 1.0. Bundled; weekly rolling export
`ilc2`	ISKO Italy (C. Gnoli and contributors)	2019-10-17	10,117	Faceted, ordered by integrative levels; 9,174 concepts and 943 facet indicators	Unsettled: CC BY 3.0 in the SKOS file, CC BY-SA 3.0 in BARTOC, no site page states a licence. Not bundled
`arxiv`	arXiv (arxiv-base)	2024-04-01	192	8 groups, 38 archives (18 retired), categories	MIT (arxiv-base). Bundled
`msc`	Mathematical Reviews (AMS) and zbMATH Open	2020	6,603	63 sections, subsections, leaf codes	CC BY-NC-SA 4.0. Not bundled (built from the official CSV)
`jel`	American Economic Association	2026	1,015	20 categories, subcategories, codes (A, A1, A10)	All rights reserved; terms grant personal use only. Not bundled
`acm-ccs`	ACM	2012	2,113	13 top-level concepts, depth up to 6; 159 concepts have more than one parent	ACM copyright; "freely available for educational and research purposes". No redistribution grant. Not bundled
`ipc`	WIPO	2026.01	80,145	8 sections, 132 classes, 655 subclasses, 7,668 main groups, 71,682 subgroups	CC BY 4.0 (WIPO Open Access Policy). File too large to commit; built by a pinned builder
`openalex-topics`	OpenAlex	2026-10-09	4,798	4 domains, 26 fields, 252 subfields, 4,516 topics; citation-derived, LLM-named	CC0 1.0. Bundled
`unesco-thesaurus`	UNESCO	2026-08-31	4,595	7 domains, 88 microthesauri, about 4,500 concepts; 229 concepts have 2 to 6 broader terms	CC BY-SA 3.0 IGO. Adaptation stays share-alike
`eurovoc`	Publications Office of the EU	4.24 (2026-07-08)	7,663	21 domains, 127 microthesauri, 7,515 descriptors; 8 levels	Sources disagree (Commission reuse notice, CC BY 4.0 for editorial content, CC0 in BARTOC). Not bundled
`DDC`	OCLC	ed. 23	n/a	Decimal tree; 1,000 classes at three digits	Copyright and trademark OCLC; full schedules only through paid WebDewey
`UDC`	UDC Consortium	Summary	n/a	Faceted decimal; Summary has about 2,600 of 70,000+ entries	Full reference file licensed to restricted communities; Summary CC BY-NC 4.0 from 2026 (see text)

2 A SURVEY OF THE CLASSIFICATION SYSTEMS

This section describes the thirteen scheme builds that the benchmark scores, and gives DDC and UDC as context. The benchmark uses twelve distinct schemes: LCC enters twice, as the published outline (`lcc`) and as the complete schedules (`lcc-full`). Facts about structure and size are those recorded by the scheme builders in the repository. Facts about licences are those recorded in the scheme READMEs and in the licensing report of the project (checked 9–10 October 2026); they are a description of what each publisher states, not legal advice.

2.1 Library classifications: LCC, DDC, UDC

LCC is maintained by the Library of Congress. Its schedules are published as PDFs for letters A–Z [64], and the Library describes its linked-data service as a public-domain data set [65]. The outline has 21 main classes (the letters A–Z less I, O, W, X and Y). The complete schedules are not available as one machine-readable current file: the project's report found a 2016 MARC bulk snapshot of 649,332 records, about 2.5 million per-URI records on id.loc.gov, and the PDFs, which the `lcc-full` builder parses. Whether LCC is free to redistribute depends on jurisdiction. US government works are not subject to copyright [66], but LC's Classification Web notice says LC-originated classification data are copyrighted except within the USA [67]. LCC has been developed in successive editions since the early twentieth century [68].

DDC is owned by OCLC [17]. Full schedules are available through the subscription WebDewey, and OCLC's terms for Dewey linked data prohibit scraping, indexing and long-term storage; the base set is free to view only. The DDC 22 Summaries PDF carries "All rights reserved". DDC is therefore absent from the benchmark, although OCLC describes it as the world's most widely used way to organize library collections [17]. The FDC was written from the out-of-copyright 1922 eleventh edition of DDC and checked only for compatibility.

UDC is maintained by the UDC Consortium. The full Master Reference File is licensed, with several licence types for restricted communities, and embedding it in software requires a paid publishing licence [18]. The licensing report records the UDC Summary (about 2,600 of 70,000+ entries) as CC BY-NC 4.0 from 2026, where earlier search snippets and the survey report described a share-alike Creative Commons 3.0 licence. We use the later statement. Either way the non-commercial condition, or the share-alike condition, blocks use inside a permissively licensed tool.

2.2 Open universal and near-universal schemes

No open-licensed scheme is at once universal, deep and equipped with a full LCC or DDC crosswalk. The open candidates are shallow, or are not shelf orders. The survey report records the main ones as follows.

BK (Basisklassifikation) is a hierarchical decimal subject classification for scholarly works, maintained by the Gemeinsamer Bibliotheksverbund [19]. It has 2,140 classes in this export (BARTOC lists 2,139, and the export carries one class marked as deleted). Its data are CC0 and refreshed weekly. It is academically biased, and its crosswalk is thin: the survey counted 130 mappings from BK in coli-conc, of which 49 point to DDC edition 23 and none to LCC. **FDC** is CC0 and DDC-shaped, but frozen since 2010 [20]. **ILC2** [21] is universal and faceted, orders phenomena by integrative levels rather than by library convention, and has no settled licence. Its SKOS file declares CC BY 3.0 while the BARTOC record lists CC BY-SA 3.0, and no page on the maintainers' site states a licence; the project therefore treats the data as CC BY-SA 3.0 if redistributed.

OpenAlex Topics is a four-level, citation-derived and LLM-named hierarchy for research literature only, released as CC0 [22], [53], [69]. **UNESCO Thesaurus** and **EuroVoc** are thesauri, narrower than a universal scheme, with a domain–microthesaurus–descriptor layout. **IPC** is the International Patent Classification, a deep scheme that is relevant here as a large hierarchy over a technical domain. WIPO's open access policy issues its online content under CC BY 4.0 unless marked otherwise, and the master files carry no other notice [70], [71]. The remaining schemes are domain schemes: arXiv's category taxonomy [72], MSC2020 [73], JEL [74] and ACM CCS [75]. Of these, only arXiv's carries an open software licence, and JEL publishes no open licence.

ACM CCS, 1998 and 2012. The 2012 system replaced the 1998 edition. ACM says it mapped the old scheme to the new one and shows both terms on article citation pages, but publishes no 1998-to-2012 mapping file; its public download contains the 2012 tree only. We found no mapping file. This matters below, because the arXiv `acm-class` field of the corpus carries CCS 1998 codes (Section 4).

Other surveyed resources. The survey also recorded resources that are supplements rather than shelf orders: Wikidata (CC0, a class graph, not a shelf order), FAST (ODC-By headings), LCSH (headings with no class numbers), GND and Thema. Wikidata is CC0 and holds 942 items with both a DDC and an LCC statement, the only open DDC–LCC concordance the survey measured, and a small one. The Open Shelves Classification was never completed. The survey found no open machine-readable copy of the Colon, Bliss or ICC classifications.

3 METHOD

3.1 Similarity graph

Each item's text is the title and abstract joined by a full stop (titles only for the book holdout, which has no descriptions). It is embedded with `sentence-transformers/all-MiniLM-L6-v2` at a pinned ONNX revision [76], [77], [78], [79], and vectors are L2-normalized. Raw cosine similarities between unrelated texts are not centred on zero, so we subtract the exact mean cosine τ over all unordered pairs: $\tau = 0.109$ on the arXiv corpus and $\tau = 0.173$ on the holdout. Each item keeps its $k = 20$ most similar items. The weight of an edge is $w_{ij} = \max\big(0, \cos_{ij} - \tau\big)$. The graph is symmetrized by taking the maximum, and zero-weight edges are dropped. That leaves 148,111 edges on the corpus and 7,297 on the holdout. Every scheme and the reference tree are scored on this one graph, so all comparisons on a corpus are paired. The classifier's optional embedding tier uses a different model (BGE-small, not used in this run). This follows the benchmark design, which aims to avoid a circular advantage for schemes whose labels embed well.

3.2 From a scheme and placements to a tree

A scheme is a set of classes, each with at most one parent, so it is a tree or a forest. For polyhierarchies such as EuroVoc and the UNESCO Thesaurus, the port has already chosen one primary parent per concept. Given one placement per item, we keep the used classes and all their ancestors. Forest roots hang under a virtual root. Each item becomes a leaf under its class node, including when that class is an internal node. An item with no placement becomes a leaf directly under the root, so every pair involving it pays the full corpus size. Coverage is therefore penalized *inside* the cost, and we also report it separately. All reported runs use one class per item: the first listed code for gold labels and the classifier's top choice otherwise. The harness also supports fractional (multi-copy) placement, which we leave for later work.

3.3 Cost, revenue and skill

Let n be the number of items (placed or not), E the edges of the graph, $W = \sum_{(i,j)\in E} w_{ij}$, and $N(u)$ the number of items in the subtree of node u. Then

$$\begin{aligned} \mathrm{cost}(T) &= \sum_{(i,j)\in E} w_{ij} N(\mathrm{lca}_T(i,j)), \\ \mathrm{cost}_{\mathrm{norm}} &= \frac{\mathrm{cost}(T)}{nW}, \\ \mathrm{revenue}_{\mathrm{norm}} &= 1 - \mathrm{cost}_{\mathrm{norm}}. \end{aligned} \tag{1}$$

A tree with every item under the root has $\mathrm{cost}_{\mathrm{norm}} = 1$. By the Moseley–Wang identity, $nW - \mathrm{cost}(T)$ is the revenue, so $\mathrm{revenue}_{\mathrm{norm}}$ is revenue over its trivial upper bound. The sum runs over graph edges only. This is a kNN-sparsified Dasgupta cost: the pairs that matter are those the embedding calls close.

Revenue depends on the corpus and the graph, so we normalize it against a reference tree T_{ref}. That tree is a deterministic average-linkage (UPGMA) dendrogram on the same graph, with missing edges treated as similarity 0:

$$\mathrm{skill}(T) = \frac{1 - \mathrm{cost}_{\mathrm{norm}}(T)}{1 - \mathrm{cost}_{\mathrm{norm}}(T_{\mathrm{ref}})}. \tag{2}$$

Skill is 0 for the flat tree and 1 for a tree as good as average linkage. Average linkage is a $\frac{1}{3}$-approximation of

maximum revenue [32], so the reference is an achievable tree, not an optimum. Skill may exceed 1. The reference reaches $\text{cost}_{\text{norm}} = 0.152$ on the corpus and 0.301 on the holdout.

The reference is a binary tree over individual items. A scheme's tree stops at classes, and all items inside one class share an LCA. This tie is intended: it models scanning the class shelf. It also means that no scheme whose classes hold more than one item can reach skill 1 easily. Skill compares a scheme with a data-fitted tree. It is not a pass mark.

3.4 Gold and shelfmark-placed subsets

A score depends on the scheme *and* on how works were placed into it. We keep two subsets apart.

- **Gold (track A).** Labels supplied by the source: arXiv primary categories, author-supplied MSC and JEL codes on the arXiv corpus, and LC call numbers on the book holdout. Call numbers are resolved to the deepest covering class of the bundled LCC outline (`lcc`) and of the complete LCC schedules (`lcc-full`). The resolver tries the full call number, then the class number without cutters, then the class letters.
- **Placed (track P).** Labels assigned by shelfmark's lexical classifier (`profile:lexical@2`). Each scheme gets an isolated store with no hand-written rules. Placements below the classifier's 0.8 confidence target are kept and flagged. Fallback placements, where no tier produced a candidate, count as unplaced.

The two subsets are never ranked against each other. Where both exist for the same scheme and items (arXiv, MSC and JEL on the corpus; LCC and `lcc-full` on the holdout), their difference measures the classifier's effect (Section 5.3).

3.5 Structure, coverage and navigation

For each run we report:

- **Coverage:** the share of items with a placement.
- **Depth:** the mean, median, 95th percentile and maximum item depth, in edges from the root to the item leaf. The root is depth 0, and an item placed in a top-level class has depth 2.
- **Branching:** the fan-out of used internal nodes. Item leaves count as children, so unplaced items inflate root fan-out.
- **Balance:** the effective number of classes $\exp(H)$ and the Gini coefficient. Both use the distribution $p_u \propto N(u)$ over used class nodes at *all* levels, so an item counts once at each ancestor class. These are multi-level summaries, not leaf-partition statistics.
- **Expected navigation cost** under uniform use, with two menu models. For an item i with ancestors u of fan-out $b(u) > 1$,

$$\text{nav}_{\log}(a) = \frac{1}{n} \sum_i \sum_u (a + \log_2 b(u)) \qquad (3)$$

is a Hick–Hyman-style decision cost with per-level overhead $a \in \{0, 1, 2\}$. The second model is

$$\text{nav}_{\text{scan}} = \frac{1}{n} \sum_i \left[\sum_u \frac{b(u)+1}{2} + \frac{n_c+1}{2} \right], \qquad (4)$$

a linear menu scan plus a scan of the n_c items under the item's deepest class. These parameters are assumptions, not fitted values. We report the sweep over a.

3.6 Implementation

The harness is a Rust crate, `shelfmark-bench`. It walks parent pointers to find LCAs, which the shallow class trees make cheap, and computes the average-linkage reference once per graph with a size-weighted (Lance–Williams) update and deterministic tie-breaking. Unit tests check a hand-computed cost, the identity cost + revenue $= nW$, that a root-only tree has $\text{cost}_{\text{norm}} = 1$, and that the embedder, the graph and the reference are deterministic. Each result records the SHA-256 of the graph, the induced tree and the scheme file.

4 CORPUS

arXiv corpus. The arXiv OAI-PMH endpoint (`arXivRaw`) was harvested for records with datestamps from 2026-01-01 to 2026-10-09 in eight top-level groups. A seeded SHA-256 ranking then sampled 10,000 records, proportional to group size with a floor of 100 per group: cs 3,968 (39.7%), physics 2,630 (26.3%), math 2,042 (20.4%), eess 519 (5.2%), stat 345 (3.5%), econ 185 (1.8%), q-bio 172 (1.7%) and q-fin 139 (1.4%). arXiv applies CC0 1.0 to metadata [80]. Datestamps follow record updates, so older papers appear: 69.7% of identifiers carry a 2026 prefix. Computer science, mathematics and physics make up 86.4% of the sample. Gold labels in other schemes are sparse. In the corpus file used for the original runs, 1,348 papers carry an MSC code, 132 an ACM class field and 2 a JEL code. A later correction of the label fields (Section 5) leaves the text, order and identifiers unchanged; it gives 1,325 papers with MSC 2020-shaped codes and 317 papers with 690 ACM CCS 1998 codes.

Open Library holdout. 500 Open Library edition records with a non-empty `lc_classifications` array were drawn through the Open Library Search and editions APIs [81]. They hold titles only, with no description: no row has text beyond the title, and titles average 3.5 words. The holdout is skewed: 382 records (76.4%) have a first call number in class P (language and literature). The other 13 letters seen are B 32, H 19, Q 8, D 5, M 4, R 3, U 2, N 2, C 2, J 1, G 1, E 1, L 1. 37 records hold only an empty string in the array. Open Library asks contributors to share under CC0 1.0 [82]. Both files are pinned by SHA-256 and by ID lists.

Schemes. Table 1 lists the 13 builds with their class counts and licence status. Five are general or universal in intent: LCC (outline and complete schedules), FDC, BK and ILC2. The rest are thesauri or vocabularies (UNESCO Thesaurus, EuroVoc, OpenAlex Topics) or domain schemes (arXiv, MSC, JEL, ACM CCS, IPC). Each

Table 2. Gold placements on the arXiv corpus. The `msc` row is the corrected gold run (see text); `arxiv` and `jel` are from the original run.

scheme	cov.	cost_norm	revenue_norm	skill	eff. classes	mean depth	max depth	max fan-out	Gini	nav_log ($a = 1$)
arxiv	100.0%	0.333	0.667	0.786	36.5	3.90	4	797	0.795	18.3
msc	13.2%	0.984	0.016	0.018	461.5	1.39	4	25	0.599	15.2
jel	0.02%	1.000	0.000	0.000	6.0	1.00	4	1	0.000	14.3

Table 3. Gold placements on the book holdout. LC call numbers are resolved to the deepest covering class.

scheme	cov.	cost_norm	revenue_norm	skill	eff. classes	mean depth	max depth	max fan-out	Gini	nav_log ($a = 1$)
lcc-full	92.2%	0.706	0.294	0.421	85.2	5.19	10	68	0.718	19.7
lcc	92.2%	0.708	0.292	0.419	35.0	4.20	9	89	0.791	18.6

Table 4. Placed by shelfmark (`profile:lexical@2`) on the arXiv corpus.

scheme	cov.	cost_norm	revenue_norm	skill	eff. classes	mean depth	max depth	max fan-out	Gini	nav_log ($a = 1$)
openalex-topics	97.3%	0.641	0.359	0.423	77.6	4.45	5	554	0.892	26.9
unesco-thesaurus	97.3%	0.721	0.279	0.329	146.7	4.90	9	535	0.852	26.7
ipc	98.0%	0.769	0.231	0.272	317.8	6.31	10	794	0.838	27.3
ilc2	100.0%	0.819	0.181	0.214	295.1	5.09	13	417	0.824	23.4
lcc-full	100.0%	0.823	0.177	0.209	657.2	6.06	11	121	0.803	22.9
acm-ccs	74.5%	0.831	0.169	0.200	84.9	3.61	7	877	0.861	23.7
fdc	77.0%	0.834	0.166	0.196	54.5	3.21	7	1103	0.836	25.6
eurovoc	97.4%	0.848	0.152	0.179	269.3	4.98	8	350	0.821	24.8
lcc	87.7%	0.849	0.151	0.178	129.0	4.08	10	661	0.837	25.4
arxiv	42.6%	0.891	0.109	0.128	14.5	2.21	4	1665	0.856	20.3
bk	48.1%	0.902	0.098	0.116	75.0	3.15	8	788	0.841	20.9
jel	47.9%	0.903	0.097	0.115	28.9	2.27	4	972	0.806	19.4
msc	39.2%	0.950	0.050	0.059	114.4	2.00	4	380	0.800	17.8

Table 5. Placed by shelfmark on the book holdout (general schemes only).

scheme	cov.	cost_norm	revenue_norm	skill	eff. classes	mean depth	max depth	max fan-out	Gini	nav_log ($a = 1$)
lcc-full	56.6%	0.925	0.075	0.107	340.8	4.14	11	13	0.492	14.0
ilc2	45.6%	0.954	0.046	0.065	188.1	2.94	11	25	0.527	12.5
unesco-thesaurus	15.8%	0.989	0.011	0.016	93.8	1.65	7	8	0.439	10.8
lcc	8.4%	0.996	0.004	0.005	63.0	1.29	8	6	0.329	10.2
fdc	5.2%	0.998	0.002	0.003	31.1	1.14	7	4	0.314	10.1
bk	4.4%	0.999	0.001	0.002	30.8	1.23	7	7	0.439	10.1

scheme file's SHA-256 is recorded, and the rerun under the corrected classifier reused byte-identical files.

5 RESULTS

Every table in this section is generated from the committed result files (Section 10). Placed rows are sorted by $\text{cost}_{\text{norm}}$, best first. All scores are on the graphs of Section 3.1; skill is relative to the average-linkage reference.

5.1 Gold placements

arXiv's own taxonomy, applied to arXiv papers by their authors, reaches skill 0.786 with full coverage (Table 2). This is the highest score in the study. We read it as an upper reference for what a well-fitted scheme with correct placements achieves on this graph, not as evidence that arXiv's taxonomy is a good *universal* scheme. The arXiv categories and the abstracts come from the same authors, and the corpus was sampled by arXiv group.

On the books, LCC call numbers resolved into the outline reach skill 0.419 and into the complete schedules 0.421, at the same coverage (92.2%, Table 3). The complete schedules use more effective classes (85 against 35) and are deeper (mean depth 5.19 against 4.20). Yet cost barely moves (0.706 against 0.708). With 500 short titles and a holdout that is three quarters class P, the title-embedding graph does not reward the finer LCC distinctions. This does not mean the distinctions are useless; it is a limit of what the graph can see.

MSC and JEL gold results are uninformative. With 13.2% and 0.02% coverage, almost every pair involves an item under the root. The MSC gold run was rescored after a correction of the corpus's `msc` field (it had held markers and words from the `msc-class` text) and of the MSC normalizer, which now accepts section and form codes such as `65-02`. Coverage moved from 13.1% to 13.2% and skill from 0.0183 to 0.0185. The corrected run is reported here. ACM CCS still has no gold run. The corpus's `acm_ccs` field originally held the MSC-style `msc-class` text copied from arXiv's ACM-class field. After the correction it holds 690 ACM CCS 1998 codes such as `I.2.6` on 317 papers, but the scheme is CCS 2012, whose codes are paths of 8-digit concept IDs, and no 1998-to-2012 crosswalk is available (Section 2). The run log records the run as skipped for that reason.

Table 6. Placement statistics for the `lexical@2` runs. "Flagged" placements are below the 0.8 confidence target but kept. "Unplaced" counts fallbacks. "Top matched term" is the vocabulary term that sent items to the top class.

scheme	set	placed	flagged (<0.8)	unplaced	top class	top share	top matched term
acm-ccs	corpus	7,445	6,854	2,555	10002944.10011123.10011674	11.8%	performance
arxiv	corpus	4,260	3,325	5,740	cs.PF	39.1%	performance
bk	corpus	4,806	3,980	5,194	01.00	16.4%	general
eurovoc	corpus	9,737	9,582	263	C2228	3.6%	paper
fdc	corpus	7,696	7,023	2,304	676	14.3%	paper
ilc2	corpus	9,999	9,999	1	edqam	4.2%	s
ipc	corpus	9,799	9,725	201	G09B23/00	8.1%	models
jel	corpus	4,786	3,854	5,214	M53	20.3%	training
lcc	corpus	8,765	8,320	1,235	QA299.6-433	7.5%	analysis
lcc-full	corpus	9,999	9,999	1	QC243.3.E5	1.2%	energy
msc	corpus	3,916	2,999	6,084	51-XX	9.7%	geometry
openalex-topics	corpus	9,732	9,616	268	F21	5.7%	energy
unesco-thesaurus	corpus	9,731	9,529	269	c155	5.5%	time
bk	holdout	22	22	478	71.32	31.8%	man
fdc	holdout	26	26	474	750	11.5%	art
ilc2	holdout	228	226	272	edqam	11.0%	s
lcc	holdout	42	42	458	QH501-531	7.1%	life
lcc-full	holdout	283	283	217	K23	2.1%	s
unesco-thesaurus	holdout	79	77	421	c155	5.1%	time

5.2 Placement by the lexical classifier

On the corpus, skill under classifier placement runs from 0.059 (msc) to 0.423 (openalex-topics). Second is unesco-thesaurus at 0.329. Coverage explains much of the order: across the 13 schemes the Spearman correlation between coverage and skill is 0.76 (Figure 1). Among the 6 schemes with at least 95% coverage, skill still spreads from 0.179 to 0.423, so coverage does not explain everything. Within that group, OpenAlex Topics scores best. It was built from research literature, the same kind of text as the corpus [22]. EuroVoc, a policy and legal thesaurus, scores lowest. That fits the expectation that a vocabulary matched to the corpus domain organizes it better. But Table 6 shows the classifier produces many low-confidence placements: in 9 of the 13 corpus runs, over 90% of placed items are flagged. The classifier's reach is itself part of the score (Section 5.3).

The holdout results are near zero for every scheme except `lcc-full` and `ilc2` (Table 5). On titles of 3.5 words, the lexical classifier finds a candidate for only 4.4% (BK) to 56.6% (complete LCC) of the books.

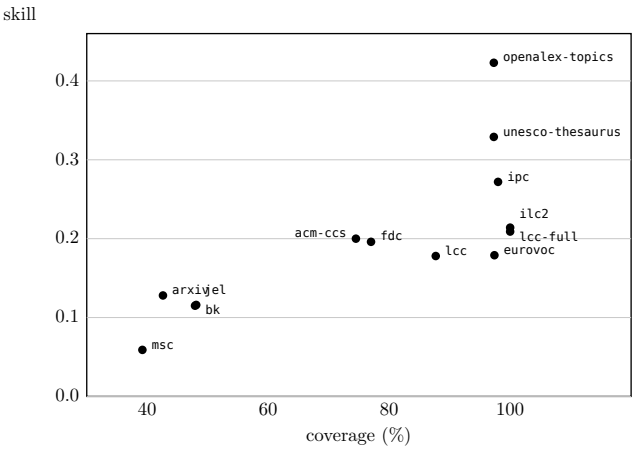


Figure 1. Coverage and skill for the 13 schemes placed by the lexical classifier on the arXiv corpus (data of Table 4).

Table 7. Gold against classifier placement on identical items and graph.

scheme	set	gold cov.	placed cov.	gold skill	placed skill	placed − gold
arxiv	corpus	100.0%	42.6%	0.786	0.128	−0.658
msc	corpus	13.2%	39.2%	0.018	0.059	+0.041
jel	corpus	0.02%	47.9%	0.000	0.115	+0.115
lcc	holdout	92.2%	8.4%	0.419	0.005	−0.413
lcc-full	holdout	92.2%	56.6%	0.421	0.107	−0.314

5.3 How large is the classifier effect?

Table 7 compares gold and classifier placements for the same scheme on the same items and graph.

Where gold coverage is high (arXiv, LCC), the classifier loses most of the skill. For arXiv, skill falls from 0.786 to 0.128. The classifier places only 42.6% of the papers and sends 39.1% of those to `cs.PF` (Performance) on the word "performance" (Table 6). For the LCC outline on books, skill falls from 0.419 to 0.005. Where gold coverage is tiny (MSC, JEL), the classifier scores *higher* than the gold labels, only because it places more items. Neither direction measures scheme quality. In this run, the classifier, not the scheme, accounts for most of the differences in classifier-placed skill. The ranking in Table 4 is therefore a ranking of *scheme as reached by this classifier*. We do not present it as a ranking of schemes.

5.4 Navigation

The navigation models answer a different question from cost: how much reading it takes to reach an item, not whether related items are near each other. With unplaced items in the root menu, the two models diverge. $\mathrm{nav}_{\log}$ *drops* as coverage falls, because one large root menu costs only $\log_2 b$. $\mathrm{nav}_{\mathrm{scan}}$ rises steeply, because the root must be scanned linearly. Among high-coverage runs, the complete LCC schedules have the lowest scan cost ($\mathrm{nav}_{\mathrm{scan}}$ 52 on the corpus) and the most effective classes. This is the

Table 8. Depth and navigation cost under uniform use (items in the root menu included). a is the per-level overhead of Eq. 3.

scheme	labels	set	p50 depth	p95 depth	root fan-out	nav_log $a=0$	$a=1$	$a=2$	nav_scan
arxiv	gold	corpus	4	4	8	15.2	18.3	21.5	291
msc	gold	corpus	1	4	8738	13.9	15.2	16.5	4372
jel	gold	corpus	1	1	10000	13.3	14.3	15.3	5001
lcc	gold	holdout	4	6	53	14.8	18.6	22.4	69
lcc-full	gold	holdout	5	7	53	15.4	19.7	24.1	56
acm-ccs	placed	corpus	4	6	2568	20.5	23.7	27.0	1476
arxiv	placed	corpus	1	4	5748	18.5	20.3	22.2	3231
bk	placed	corpus	1	6	5199	18.3	20.9	23.5	2736
eurovoc	placed	corpus	5	6	284	20.2	24.8	29.4	269
fdc	placed	corpus	4	5	2314	22.4	25.6	28.7	1413
ilc2	placed	corpus	5	7	68	18.9	23.4	27.9	136
ipc	placed	corpus	6	9	209	22.3	27.3	32.3	322
jel	placed	corpus	1	4	5231	17.4	19.4	21.4	2876
lcc	placed	corpus	4	6	1255	21.7	25.4	29.0	825
lcc-full	placed	corpus	6	8	22	17.5	22.9	28.2	52
msc	placed	corpus	1	4	6141	16.0	17.8	19.7	3128
openalex-topics	placed	corpus	5	5	272	22.4	26.9	31.3	311
unesco-thesaurus	placed	corpus	5	7	276	22.0	26.7	31.4	305
bk	placed	holdout	1	1	481	9.1	10.1	11.2	242
fdc	placed	holdout	1	2	482	9.0	10.1	11.2	243
ilc2	placed	holdout	1	7	300	10.3	12.5	14.6	155
lcc	placed	holdout	1	4	470	9.1	10.2	11.3	237
lcc-full	placed	holdout	5	9	236	11.3	14.0	16.7	125
unesco-thesaurus	placed	holdout	1	6	427	9.4	10.8	12.2	216

Table 9. Effect of the stopword fix (before: `lexical@1`; after: `lexical@2`). "Top class" gives the largest single-class share of placed items and its code.

scheme	set	top class before	top class after	coverage	cost_norm	skill	Δ skill
acm-ccs	corpus	11.8% 10002944.10011123.10011674	11.8% 10002944.10011123.10011674	74.5% → 74.5%	0.831 → 0.831	0.200 → 0.200	0.000
arxiv	corpus	39.1% cs.PF	39.1% cs.PF	42.6% → 42.6%	0.891 → 0.891	0.128 → 0.128	0.000
bk	corpus	16.4% 01.00	16.4% 01.00	48.1% → 48.1%	0.902 → 0.902	0.116 → 0.116	0.000
eurovoc	corpus	3.6% C2228	3.6% C2228	97.4% → 97.4%	0.848 → 0.848	0.179 → 0.179	+0.001
fdc	corpus	14.3% 676	14.3% 676	77.0% → 77.0%	0.834 → 0.834	0.196 → 0.196	0.000
ilc2	corpus	73.7% agy993	4.2% edqam	100.0% → 100.0%	0.825 → 0.819	0.206 → 0.214	+0.007
ipc	corpus	60.1% C01G	8.1% G09B23/00	99.4% → 98.0%	0.781 → 0.769	0.259 → 0.272	+0.014
jel	corpus	18.7% M53	20.3% M53	51.7% → 47.9%	0.902 → 0.903	0.116 → 0.115	−0.001
lcc	corpus	7.5% QA299.6-433	7.5% QA299.6-433	88.4% → 87.7%	0.848 → 0.849	0.179 → 0.178	0.000
lcc-full	corpus	22.1% PL2813	1.2% QC243.3.E5	100.0% → 100.0%	0.800 → 0.823	0.236 → 0.209	−0.027
msc	corpus	9.7% 51-XX	9.7% 51-XX	39.2% → 39.2%	0.950 → 0.950	0.059 → 0.059	0.000
openalex-topics	corpus	5.7% F21	5.7% F21	97.4% → 97.3%	0.641 → 0.641	0.423 → 0.423	0.000
unesco-thesaurus	corpus	17.8% c493	5.5% c155	98.9% → 97.3%	0.732 → 0.721	0.316 → 0.329	+0.013
bk	holdout	31.8% 71.32	31.8% 71.32	4.4% → 4.4%	0.999 → 0.999	0.002 → 0.002	0.000
fdc	holdout	11.5% 750	11.5% 750	5.2% → 5.2%	0.998 → 0.998	0.003 → 0.003	0.000
ilc2	holdout	13.7% agqs99	11.0% edqam	56.8% → 45.6%	0.934 → 0.954	0.095 → 0.065	−0.029
lcc	holdout	7.1% QH501-531	7.1% QH501-531	8.4% → 8.4%	0.996 → 0.996	0.005 → 0.005	0.000
lcc-full	holdout	3.9% BQ940	2.1% K23	61.0% → 56.6%	0.917 → 0.925	0.119 → 0.107	−0.011
unesco-thesaurus	holdout	5.0% c155	5.1% c155	16.0% → 15.8%	0.989 → 0.989	0.016 → 0.016	0.000

deep, narrow-shelf trade-off: more levels to descend, but short shelves at the bottom. The models' parameters are assumptions, so Table 8 shows the trade-off and does not settle it. The menu studies cited in Section 1.1 disagree on the best depth and breadth [45], [47], [48], so they do not fix the parameters either.

6 A METHODS NOTE: THE STOPWORD DEFECT

The first placed run (`profile:lexical@1`) showed single-class captures. The classifier's vocabulary tier matched complete class labels and terms, and some of them consist only of English function words. ILC2 has a class labelled "to", IPC a C01G catchword "TO" and the complete LCC a class `PL2813` labelled "To". Any abstract containing "to" matched them. The fix (classifier `vocab-2`, `profile:lexical@2`) drops normalized labels and terms whose tokens are all stopwords. Labels with at least one content word still match in full. We then reran only the placed part, on the same graphs, scheme files (byte-identical, checked by SHA-256) and reference costs. The gold results do not depend on the classifier and were not rerun on this account. Every placed number in this paper comes from the rerun.

The fix removed the captures (Table 9). ILC2′s largest class fell from 73.7% to 4.2% of placed papers, IPC's from 60.1% to 8.1%, complete LCC's from 22.1% to 1.2% and the UNESCO Thesaurus's from 17.8% to 5.5%. Skill moved much less. On the corpus, 6 of 13 schemes changed skill by at least 0.0005, and Kendall's τ_b between the skill values before and after is 0.95. Complete LCC *lost* skill (0.236 → 0.209) even though its placements became plausible. A plausible explanation, not tested here, is that

Table 10. The fitted scheme against published schemes on the same graphs. "own, in-sample" rows are the fitted assignment of the corpus it was fitted to. "own insertion" rows file works by nearest leaf centroid in the embedding space of the graph. Rows through `lexical@2` use the classifier of Section 5.2. Lower is better for cost and J.

evaluation set	scheme	placements	coverage	cost_norm	skill	J	nav_log ($a = 1$)
corpus, n = 10,000	learned	own, in-sample (fitted on this corpus)	1.000	0.172	0.976	0.264	18.5
corpus, n = 10,000	arxiv	gold (source labels)	1.000	0.333	0.786	0.425	18.3
corpus, n = 10,000	learned	shelfmark lexical@2	0.999	0.379	0.732	0.535	31.1
corpus, n = 10,000	openalex-topics	shelfmark lexical@2	0.973	0.641	0.423	0.776	26.9
corpus, n = 10,000	lcc-full	shelfmark lexical@2	1.000	0.823	0.209	0.937	22.9
held-out 20%, n = 2,000	learned (fit on the other 80%)	own insertion, held out	1.000	0.172	0.922	0.253	16.1
held-out 20%, n = 2,000	learned (full fit)	own, in-sample	1.000	0.179	0.914	0.260	16.2
held-out 20%, n = 2,000	arxiv	gold (source labels)	1.000	0.342	0.733	0.422	15.9
held-out 20%, n = 2,000	openalex-topics	shelfmark lexical@2	0.975	0.652	0.388	0.758	21.2
book holdout, n = 500	learned	own insertion (embedding nearest centroid)	1.000	0.443	0.797	0.542	19.7
book holdout, n = 500	lcc	gold (source labels)	0.922	0.708	0.419	0.801	18.6
book holdout, n = 500	lcc-full	gold (source labels)	0.922	0.706	0.421	0.804	19.7
book holdout, n = 500	lcc-full	shelfmark lexical@2	0.566	0.925	0.107	0.995	14.0
book holdout, n = 500	learned	shelfmark lexical@2	0.086	0.992	0.012	1.044	10.6

the wrong placements had piled most papers that share "to" into a single class. Within such a pile, the edges the graph keeps still connect similar papers, and the pile is far smaller than the whole corpus, so its edges cost less than they would under the root. A metric can reward a degenerate placement more than a spread of honest but noisy placements. This is a known property of a cost that charges same-class pairs the class size: one large class still beats the root. It is also why we always report largest-class share and placement statistics next to skill.

7 A FITTED ORDER-PRESERVING SCHEME

The benchmark makes a converse question natural: if a scheme can be scored, can one be *fitted* to the score, under constraints a library needs? We report a first, short experiment; the full description is in a companion draft.

Objective. On the graph of Section 3.1, the fitted scheme minimizes

$$J(T) = \mathrm{cost}_{\mathrm{norm}}(T) + \lambda N(T), \quad \lambda = 0.005, \qquad (5)$$

where $\mathrm{cost}_{\mathrm{norm}}$ is the harness's cost with works under their class and N is $\mathrm{nav}_{\log}$ with $a = 1$ (Eq. 3; a leaf class's fan-out is its number of works). The constraints are 2 to 12 child classes per class, at most 4 class levels, leaf classes of at least 5 works, and every class a contiguous run of one fixed shelf order, the alphabetic-tree constraint of optimal order-preserving codes [42], [83]. New works are filed by splitting leaf classes only, so that every existing class keeps its code, label and parent. The fit starts from the average-linkage tree, collapses it to a bounded-shape class tree by an exact dynamic program, and applies order-preserving local search; it takes about 2.5 s on the corpus. The result has 110 classes (2, 5, 16 and 87 by level). Their labels were written by hand from sheets of class terms and central titles, by the AI coding assistant that built the prototype and not by a cataloguer.

Results. In sample, the fitted scheme reaches cost 0.1722 (average linkage 0.1518, arXiv's own categories 0.333), skill 0.9760 and J 0.2644 (Table 10). Fitted on a random 80% and filed on the other 20% by insertion, it scores 0.1722 on the test works, against 0.1793 for the full fit that saw them. Reached through the lexical classifier, it scores 0.3794: a lower cost than every published scheme's placed row, and a higher cost than arXiv's gold labels. Inserting the 500 books as new works produced 16 classes and changed no existing identifier; on the holdout graph the inserted scheme scores 0.4432 (skill 0.7971), against 0.708 for LCC call numbers. Through the classifier it places only 0.086 of the books and scores 0.9915.

Caveats. These numbers flatter the method, and we do not read them as a win over any published scheme.

- *In sample.* The corpus rows, and the classifier rows of the fitted scheme, use a tree or a vocabulary fitted on those same works. The fitted scheme's vocabulary is the corpus's own most distinctive terms.
- *Embedding circularity.* Insertion files a new work at the nearest leaf centroid in the same all-MiniLM-L6-v2 space from which the graph is built, so a new work is placed by exactly the geometry it is later scored on. This holds for the held-out rows and, more strongly, for the book holdout, where graph and insertion see only title embeddings. The classifier rows are the comparison without it. A fair test needs a second, independent similarity.
- *One corpus.* The scheme has seen only arXiv papers. Its top level splits computing from mathematics and the physical sciences, and it has no class for the humanities until insertion creates a weak one. Labels of inserted classes come from frequent title words and are not usable as captions.
- *The objective.* Dasgupta cost favours a binary split near the top, so the fitted root has two children. Library schemes have wide top levels for reasons of browsing that J does not model.
- *Weights are modelling choices.* λ, a and the depth bound are not fitted. A sweep over depth bounds 3, 4 and 6 and λ from 0.001 to 0.05 moves $\mathrm{cost}_{\mathrm{norm}}$ from 0.1577 to 0.2075 and the class count from 330 to 33.

8 DISCUSSION

What the measurement supports. The method gives one scale for schemes of different size, depth and domain. It costs little: the average-linkage reference took 58.4 s on the corpus graph and 0.1 s on the holdout graph, and the whole first run, including embedding and scheme builds, took about 45 minutes on 8 CPUs, with placement into the 416,186-class `lcc-full` the longest single job. Its inputs are pinned by hash. On gold labels it separates a well-fitted scheme (arXiv on arXiv) from a general scheme on books (LCC on the holdout) and from schemes with too little gold to judge (MSC, JEL).

What it does not yet support. It does not support a ranking of universal schemes. The general schemes that matter most for libraries (LCC, BK, FDC, ILC2) have gold labels only for LCC, and only on a small title-only holdout. Classifier placement, the only way to reach the other schemes, moves skill by as much as the differences we would want to detect (Table 7). The stopword defect shows a second risk. A classifier defect can change placements drastically while skill barely moves, or moves it the wrong way (Section 6).

Toward a fair comparison. The benchmark design lays out the controls needed. These are a cost–resolution curve against the reference cut at matched class counts; a permutation null that keeps each scheme's shape and class-size histogram; item bootstrap intervals; a label-intrinsic track that needs no works; and a noise-injection curve that calibrates classifier error on LCC, where both tracks exist. None of these was run for this draft. The largest gain would come from a large gold set of books that carry both LCC and DDC. The benchmark design counted about 3.91 million Open Library search documents with both labels and `language:eng` (queried 9 October 2026). DDC could be scored by its digit tree alone, which needs no captions.

Interpretation of skill. Skill compares a scheme with a tree fitted to the same similarity graph. A scheme encodes many relations besides textual similarity: discipline, form, audience, place, period. A low skill can mean the scheme organizes by something the embedding does not see. Our score measures agreement with one notion of relatedness. It does not measure quality in every sense that matters to libraries.

On the fitted scheme. A scheme fitted to the benchmark's own objective scores well on that objective, as it must. The result is useful as a reference point, not as a ranking: a hand-built scheme can be described as a given distance from a fitted scheme with the same number of levels on the same graph. Whether an open, learned, order-preserving classification can become a general one is open (Section 7).

9 LIMITATIONS

- **Classifier bias.** Every non-gold placement comes from shelfmark's own lexical classifier. It matches class labels and attached vocabulary, so a scheme with rich vocabulary is reached better than one with bare labels, independent of its structure. Most placements are below the classifier's own confidence target (Table 6). Table 7 shows the classifier effect is as large as the effects we want to measure. No other classifier (for example Annif or an LLM) was run for comparison.
- **Science-heavy corpus.** 86.4% of the corpus is computer science, mathematics or physics. Humanities, law, arts and general nonfiction appear only in the 500-record holdout, which is itself 76.4% class P. Domain schemes (arXiv, MSC, ACM CCS) are scored on their home ground and general schemes are not, so no cross-type comparison is fair.
- **Single embedding.** The graph is built from one embedding model, one value of k and one mean offset, with no sensitivity runs. A different embedding, or a similarity from citations or subject headings, could reorder results. The fitted scheme of Section 7 is the most exposed to this circularity.
- **Low gold coverage for MSC and JEL.** Only 13.2% of papers carry a valid MSC code and 0.02% a JEL code. Unplaced items hang under the root, so these gold scores measure coverage, not structure. A covered-subset score, restricted to labelled items and their edges, is not yet implemented.
- **ACM CCS gold missing.** The corpus carries CCS 1998 codes, the scheme is CCS 2012, and no official crosswalk exists (Section 2). The gold run was skipped, and we did not build a mapping by hand.
- **Scheme licensing.** Several schemes cannot be redistributed: ACM CCS (educational and research use only), JEL (no open licence), MSC (non-commercial, share-alike), EuroVoc (terms stated inconsistently) and ILC2 (conflicting licence statements) (Table 1). Their files are built locally from the publishers' downloads and verified by hash, but are not published. Exact reproduction therefore depends on the upstream downloads staying available and unchanged. DDC, which OCLC describes as the world's most widely used way to organize library collections, is absent because OCLC holds its copyright [17]. Even LCC's open status depends on jurisdiction (Section 2).
- **Titles-only holdout.** The book holdout has no descriptions. Its similarity graph is built from titles averaging 3.5 words, and the classifier finds candidates for few of them. The holdout is also small ($n = 500$), and no confidence intervals are reported.
- **Method limits.** The kNN-sparsified cost has not been checked against the dense cost. Polyhierarchies are reduced to one parent per class. Only one placement per item is used, although 35.5% of a 200-work Open Library sample of English documents had more than one LCC value. The balance statistics count each item at every ancestor class (Section 3.5). The navigation models are unfitted.
- **Self-embedding circularity of the fitted scheme.** Its insertion and the graph share an embedding, its corpus rows are in sample, and its labels were

written by the AI coding assistant that built it and have not been evaluated by people (Section 7).

- **Search for prior work.** The novelty claim rests on the search in project report 8: 94 Crossref queries over the JASIS/JASIST family, Knowledge Organization, Cataloging & Classification Quarterly, ISKO, SIG/CR and NASKO proceedings, JCDL and TPDL/ECDL, plus five web searches. It screened titles and abstracts only. Full texts and non-English literature were not searched, one person screened, and Crossref's coverage of proceedings is incomplete (ISKO proceedings before 2012 are missing).

10 REPRODUCIBILITY

All inputs and outputs are in the project repository or pinned by hash.

- **Corpus.** `bench/corpus/corpus.jsonl` and `holdout.jsonl`, with ID lists and SHA-256 hashes. Metadata only, CC0 1.0. The original runs used a `corpus.jsonl` with SHA-256 prefix `7685685d`; the corrected label fields used for the MSC gold rescoring give prefix `1027bf9d`. Neither change touches row order, identifier, title or abstract, the only inputs to the graph.
- **Similarity graphs.** Published gzipped, with manifests, under `bench/results/2026-10-10/graphs/`. The manifests give the model revision `d13954661f83248295ba75c1ed411eef3b7b936e`, the `model.onnx` SHA-256, τ, k and the graph SHA-256 (corpus graph prefix `759d3b84`).
- **Harness.** `crates/bench`, at the commit recorded in each `results.json` (`092494850adc`). The classifier for the placed run is at `f4ce24b78eda` (`profile:lexical@2`).
- **Schemes.** Bundled schemes are in the repository. Fetch-only schemes are rebuilt from the publishers' downloads, checked against committed `sources.lock.json` files, and must match the `file_sha256` recorded in `results.json`.
- **Commands.** Each result directory has a `REPRODUCE.md` listing every command. The first run took about 45 minutes and the rerun about 25 minutes, both on 8 CPUs.
- **Determinism.** With the same graph and scheme files, gold scores reproduce exactly. Placed scores reproduce exactly with the same classifier build. Rebuilding the graph from text on a different ONNX Runtime version can change near-tied neighbours, so the published graph file, not the embedding pipeline, is the reproducible artefact.
- **This paper.** `paper/method-survey/scripts/gen_tables.py` regenerates every table and every headline value (the dictionary `v` in `generated.typ`) from the result files. Table 1 and the prose of Section 2 are not generated: they are taken from the scheme READMEs and the survey and licensing reports.

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